

$$1. \begin{cases} E_1 l_1 + E_2 l_2 = \xi \\ \epsilon_1 \epsilon_0 E_1 = \epsilon_2 \epsilon_0 E_2 \end{cases}$$

$$\Rightarrow E_1 = \frac{\epsilon_2 \xi}{\epsilon_2 l_1 + \epsilon_1 l_2}, E_2 = \frac{\epsilon_1 \xi}{\epsilon_2 l_1 + \epsilon_1 l_2}$$

$$(1) W_f = D_1 = \frac{\epsilon_0 \epsilon_1 \epsilon_2 \xi}{\epsilon_2 l_1 + \epsilon_1 l_2}$$

$$(2) W_f = 0$$

$$(3) \sigma_1 E_1 = \sigma_2 E_2$$

$$E_1 = \frac{\sigma_2 \xi}{\sigma_2 l_1 + \sigma_1 l_2}, E_2 = \frac{\sigma_1 \xi}{\sigma_2 l_1 + \sigma_1 l_2}$$

$$(1) \text{ 答案变为 } W_f = \frac{\epsilon_1 \sigma_2 \xi}{\sigma_2 l_1 + \sigma_1 l_2} \text{ (上板)}, \frac{\epsilon_2 \sigma_1 \xi}{\sigma_2 l_1 + \sigma_1 l_2} \text{ (下板)}$$

$$(2) \text{ 变为 } W_f = \frac{\epsilon_2 \sigma_1 - \epsilon_1 \sigma_2}{\sigma_2 l_1 + \sigma_1 l_2} \xi$$

$$2. (1) D_{2n} - D_{1n} = \sigma_f = 0 \Leftrightarrow \epsilon_2 E_{2n} = \epsilon_1 E_{1n}$$

$$\text{同时 } E_{2t} = E_{1t}$$

$$\tan \theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{E_{1t}}{(\epsilon_1/\epsilon_2) E_{1n}} = \frac{\epsilon_2}{\epsilon_1} \tan \theta_1$$

$$\Leftrightarrow \frac{\tan \theta_2}{\tan \theta_1} = \frac{\epsilon_2}{\epsilon_1}$$

$$(2) \sigma_2 E_{2n} = \sigma_1 E_{1n}$$

$$E_{2t} = E_{1t}$$

$$\frac{\tan \theta_2}{\tan \theta_1} = \frac{E_{2t}/E_{2n}}{E_{1t}/E_{1n}} = \frac{\sigma_2}{\sigma_1}$$

$$3. (1) \rho_p = -\nabla \cdot \vec{P}$$

$$= -k \nabla \cdot \frac{\vec{r}}{r^2}$$

$$= -k \left(\frac{1}{r^2} \nabla \cdot \vec{r} + \vec{r} \cdot \nabla \left(\frac{1}{r^2} \right) \right)$$

$$= -k \left(\frac{3}{r^2} - \frac{2}{r^3} \vec{r} \cdot \vec{r} \right)$$

$$= -k \cdot \frac{1}{r^2}$$

$$\sigma_p = \rho_m - \rho_{2n}$$

$$= \rho_{1n}$$

$$= \frac{k}{r^2}$$

$$\oint \vec{P} \cdot d\vec{s} = \oint \rho_p dV$$

$$= -Q_p$$

$$= -\sigma_p$$

$$(2) \rho_p = -\nabla \cdot \vec{P}$$

$$\left\{ \begin{array}{l} \vec{D} = \epsilon \vec{E} = \vec{P} + \epsilon_0 \vec{E} = \vec{P} + \frac{\epsilon_0}{\epsilon} \vec{D} \end{array} \right.$$

$$\Rightarrow \rho_f = \nabla \cdot \vec{D} = \nabla \cdot \left(1 - \frac{\epsilon_0}{\epsilon} \right) \vec{P}$$

$$= \left(1 - \frac{\epsilon_0}{\epsilon} \right) (-\rho_p)$$

$$= k \cdot \left(1 - \frac{\epsilon_0}{\epsilon} \right) \frac{1}{r^2}$$

$$(3) \text{ 当 } r \geq R, Q_f = \int_0^R k \left(1 - \frac{\epsilon_0}{\epsilon} \right) \cdot \frac{1}{r^2} \cdot 4\pi r^2 dr$$

$$= 4\pi k \left(1 - \frac{\epsilon_0}{\epsilon} \right) R$$

$$\varphi(r) = \frac{Q_f}{4\pi \epsilon_0 r} = \frac{k \left(1 - \frac{\epsilon_0}{\epsilon} \right) R}{\epsilon_0 r}$$

$$\text{当 } 0 \leq r < R, D(r) \cdot 4\pi r^2 = \int_0^r k \left(1 - \frac{\epsilon_0}{\epsilon} \right) \cdot \frac{1}{z^2} \cdot 4\pi z^2 dz$$

$$= 4\pi k \left(1 - \frac{\epsilon_0}{\epsilon} \right) r$$

$$\Leftrightarrow \vec{D}(r) = k \left(1 - \frac{\epsilon_0}{\epsilon} \right) \frac{\vec{r}}{r^2}$$

$$\Leftrightarrow \vec{E}(r) = \frac{k}{\epsilon} \left(1 - \frac{\epsilon_0}{\epsilon} \right) \frac{\vec{r}}{r^2}$$

$$\varphi(r) = \int_r^R \frac{k}{\epsilon} \left(1 - \frac{\epsilon_0}{\epsilon} \right) \cdot \frac{1}{r} dr + \varphi(R)$$

$$= \frac{k}{\epsilon} \left(1 - \frac{\epsilon_0}{\epsilon} \right) \ln \frac{R}{r} + \frac{k \left(1 - \frac{\epsilon_0}{\epsilon} \right)}{\epsilon_0}$$

$$(4) W = \frac{1}{2} \int_V \vec{D} \cdot \vec{E} dV$$

$$= \frac{1}{2} \left(\int_0^R \frac{1}{\epsilon} k \left(1 - \frac{\epsilon_0}{\epsilon} \right)^2 \frac{1}{r^2} \cdot 4\pi r^2 dr \right)$$

$$= \frac{1}{2\epsilon} k^2 \left(1 - \frac{\epsilon_0}{\epsilon} \right) \cdot 4\pi R$$